

Model Validation Report: Probability of Default Model v1.0

Executive Summary

This validation report critically evaluates the **Probability of Default Model Version 1.0**, identifying significant conceptual and methodological issues in accordance with the **CBUAE Model Management Guidance (MMG)** and **Model Management Standards (MMS)**.

Given the identified flaws, the model presents a high risk of incorrect decision-making if deployed, necessitating a **“REJECT / DO NOT DEPLOY”** mandate.

Key Findings

1. Critical Conceptual Flaws

- **Illogical Coefficients**
 - **Variable Identified:** Income
 - **Issue:** Positive coefficient suggests that higher income increases default risk, which contradicts economic logic.
 - **Risk & Reference:** Undermines model reliability; non-compliance with CBUAE MMG Section IV on Variables' Economic Logic.
 - **Severity:**  **RED**
- **Data Leakage / Look-Ahead Bias**
 - **Variable Identified:** current_balance
 - **Issue:** Reflects post-origination behavior, introducing look-ahead bias.
 - **Risk & Reference:** Inflates AUC/Gini metrics; violates CBUAE MMG Data Integrity Clause prohibiting post-event data usage.
 - **Severity:**  **RED**
- **Multicollinearity**
 - **Variables Involved:** current_balance and loan_amount
 - **Issue:** High correlation (0.89) destabilizes model coefficients.
 - **Risk & Reference:** Coefficient inconsistency; violates CBUAE MMS Section on Model Stability.

- **Severity:**  **RED**

2. Methodological Gaps

- **Missing Weight of Evidence (WOE) Transformation**
 - **Observation:** WOE not applied.
 - **Risk & Reference:** Potential overfitting, poor discrimination; breaches CBUAE MMS on Feature Engineering.
 - **Severity:**  **AMBER**
- **Missing Out-of-Time (OOT) Tests**
 - **Observation:** No OOT validation conducted post-December 2025.
 - **Risk & Reference:** Model temporal stability untested; contravenes CBUAE MMS on Model Performance Tests.
 - **Severity:**  **RED**
- **PIT vs. TTC Adjustment**
 - **Observation:** High default rate (30.93%) without Economic Cycle Adjustment.
 - **Risk & Reference:** Over-estimation during normal periods; lacks compliance with CBUAE MMG Economic Cycle Adjustment Clause.
 - **Severity:**  **RED**
- **Calibration Test Absence**
 - **Observation:** Hosmer-Lemeshow test omitted.
 - **Risk & Reference:** Calibration unverified; CBUAE MMG Calibration Requirements unmet.
 - **Severity:**  **AMBER**

3. Quantitative Performance Review

Metric	Value
AUC (Training)	0.812
AUC (Test)	0.783
Default Rate	30.93%
Gini Coefficient	~56.6%

Note on Performance: The Gini coefficient indicates weak model discrimination when underlined by the identified conceptual flaws. The explicit omission of critical calibration tests and the absence of documented out-of-time validation denote significant methodological gaps.

4. Critical Commentary

The conflation of conceptual issues like look-ahead bias and multicollinearity with absent methodological rigour significantly compromises the model's integrity. This undermines the predictive power suggested by current metrics, questioning their practical relevance.

Conclusion

The **Probability of Default Model Version 1.0**, due to significant conceptual flaws and methodological oversights that contravene CBUAE regulatory standards, is not fit for deployment in its current form.

Immediate remediation focused on addressing data leakage, multicollinearity, and implementing rigorous validation methods (such as WOE transformations and out-of-time stability tests) is imperative for any future use consideration.

Final Mandate: ✗ REJECT / DO NOT DEPLOY